Analysis 2: HW9

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Problem 1. First denote the minimum distance from a point in U to v as d so we have $d = inf_{u \in U}\|v-u\|$ and now we need to show that there exists a sequence in U i.e (u_n) that converges to some u such that u has the smallest distance from v and u is denoted the projection of v onto U. So essentially we need to show that (u_n) is a Cauchy sequence in U and converges within the set. So note that we have,

$$\|u_m - u_n\|^2 = \|(u_m - v) - (u_n - v)\|^2 = 2\|u_m - v\|^2 + 2\|u_n - v\|^2 - \|u_m + u_n - 2v\|^2$$

We can now bound this to $\|u_m - u_n\|^2 \leq 0$ which means it Cauchy. So this means that $(u_n) \to u$ and hence there exists a limit which is the projection of v onto U. We can now easily show that if we have u and u' two limits that they need to be equal because using the parallelogram law again but with u and u'.

Now take some arbitrary v and let u_0 be the projection on U as defined above. Now we need that $v - u_0$ is perpendicular to U or that for any $u \in U^\perp$ we have $u \perp (v - u_0)$.

Now we need to show that $U \bigoplus U^\perp = H$, first clealry we know that $U \cap U^\perp = \{0\}$ because if x is in the intersection then $\langle x, x\rangle = 0$ which implies that $x = 0$. Now we need to show that $H = U + U^\perp$. So take some $v \in H$ and note that $u_0 = proj_U(v)$ then we have $v = u_0 + (v - u_0)$ where $u_0 \in U$ and $v - u_0 \in U^\perp$ as above, hence we have $v \in U + U^\perp$.